Ethan Hosier

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Education

Imperial College London — London, UK
MEng Computing - First Class Honours — *Oct. 2022 – June 2026*

Experience

Qube Research & Technologies (QRT) | *Kotlin, Python, AWS, ClickHouse, PostgreSQL* — Jun 2026–Present
Software Engineer Intern - Risk

- Architected a lock-free staging and compaction pipeline for concurrent graph updates, resolving distributed read-compute-write races while preserving consistency.
- Optimised graph compaction with parallel workers and no-op filtering and eliminated index bloat through automated staging table truncation, cutting latency from **39.25 ms to 0.047 ms**.
- Redesigned a market-risk scenario engine's temporal cache, eliminating recurring OOMs by reducing memory usage from **70 GB** across 250,000 entries to **under 200 MB** across 146 entries and enabling full historical backfills.
- Diagnosed an unbounded backpressure gap in an external library and patched its internal streaming buffer to cap queued data, preventing multi-GB memory growth and OOMs during S3 exports.

Palantir | *Python, Spark, TypeScript* — Apr 2025–Sep 2025
Forward Deployed Software Engineer, Intern

- Re-architected a Spark entity-resolution pipeline linking billions of transactions across millions of customers, parallelising customer clustering and transaction attribution, then joining results to cut execution time by **71%**.
- Optimised distributed execution through broadcast joins, dataset indexing and persisted intermediate results, selectively recomputing affected clusters to eliminate full-graph rebuilds.
- Transformed the client's financial forecasting process by developing a spreadsheet engine from scratch supporting real-time multi-user collaboration, formulas, keyboard shortcuts and configurable templates

Revolut | *Java, TypeScript, React.js, PostgreSQL, Google Cloud Platform* — Jul 2024–Sep 2024
Software Engineer, Intern

- Replaced repeated queries and full recomputation of global product configurations with an in-process cache incrementally updated from a PostgreSQL event stream, reducing query latency by **92%**.
- Reduced production pricing-rule execution time by 24% through caching and parallelising independent calculations.
- Built a dashboard for product and pricing-rule configurations, used daily by the CEO and leadership team, eliminating the need for manual SQL queries.

Projects

Distributed Web Crawler & RAG Pipeline | *Go, Redis, AWS, Terraform*

- Built a distributed crawler and RAG pipeline in Go with Redis-based coordination and autoscaling on AWS ECS/Fargate via Terraform, indexing Imperial's website in under 3 minutes using headless Chromium workers.
- Built an automated marketing platform on this infrastructure, attracting interest from 40+ companies and securing £2,500 in Imperial entrepreneurship funding and £1,000 in Microsoft Azure credits.

PintOS Operating System Kernel | *C*

- Independently implemented demand paging with clock-based page replacement, swapping, stack growth and memory-mapped files
- Added argument passing and process creation/wait handling for user programs, independently implementing system calls for process control and I/O.
- Implemented priority-based kernel thread scheduling with priority donation and a multilevel feedback queue

WACC Compiler | *Scala*

- Built a Scala compiler with parsing, type checking, scoped symbol resolution and x86-64 code generation.
- Implemented constant folding, function inlining, dead code elimination and peephole optimisations, reducing execution time by 12%

Technical Skills

Rust, C/C++, Kotlin, Java, Python, Go, TypeScript, AWS, React.js, Terraform, PostgreSQL, Haskell